AGENTIC AI – DEEP DIVE

Total Duration: 16 Hours | 10 Modules

Prerequisite Knowledge

- Working knowledge of LLMs and prompt engineering fundamentals
- Basic Python programming skills
- Familiarity with APIs and REST concepts

Target Audience

- AI/ML engineers and developers building autonomous or multi-agent systems
- Solution architects designing agent-based products and workflows
- Technical product managers exploring agentic AI use cases for their teams

Course Curriculum

Module	Topic	Details	Duration
1	**Foundations of Agentic AI**	Agents vs assistants, autonomy, the agentic loop (perceive-plan-act)	**1 Hour**
2	**Agent Architectures & Design Patterns**	ReAct, Plan-and-Execute, Reflexion, multi-agent systems	**1.5 Hours**
3	**Tool Use & Function Calling**	Tool schemas, function calling, tool-selection strategies	**1.5 Hours**
4	**Memory & Context Management**	Short-term vs long-term memory, vector stores, context window management	**1.5 Hours**
5	**Agent Frameworks**	LangGraph, AutoGen, CrewAI, Semantic Kernel, Anthropic/OpenAI Agent SDKs	**2 Hours**
6	**Multi-Agent Orchestration**	Agent communication protocols, MCP, A2A, supervisor/worker patterns	**2 Hours**

Module	Topic	Details	Duration
7	**Retrieval-Augmented Agents**	RAG pipelines within agentic workflows, hybrid search, grounding	**1.5 Hours**
8	**Guardrails, Safety & Human-in-the-loop**	Permissions, approval workflows, evaluation, failure handling	**1.5 Hours**
9	**Observability & Debugging Agents**	Tracing, logging, cost/latency monitoring, agent evaluation metrics	**1.5 Hours**
10	**Enterprise Use Cases & Capstone**	Building a customer support agent, coding agent, and ops agent	**2 Hours**